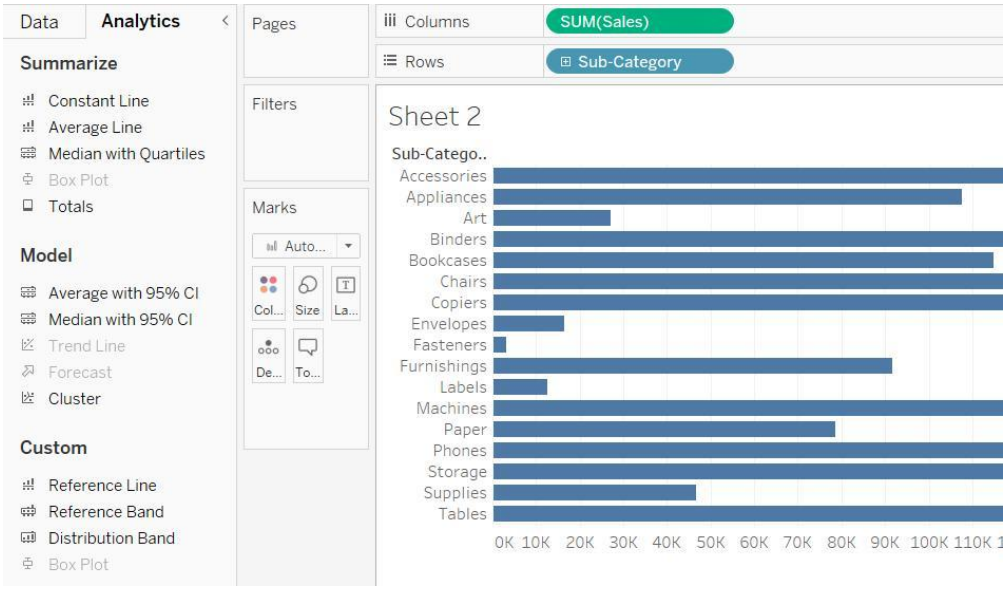
Tableau Analytics Pane — Notes

In Tableau, the Analytics pane helps you add analytical elements to a chart or visualization. Instead of showing only your data, you can use it to identify trends, compare values, find averages, and understand patterns.

Think of it like this:

- Data pane: What data do I have?
- Analytics pane: What can I learn from that data?

1. Where is the Analytics pane?

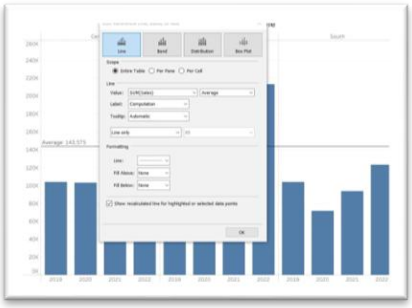


In Tableau Desktop:

1. Open a worksheet.
2. Look at the left-side panel.
3. Click Analytics, next to the Data tab.
4. Drag an analytical element onto your chart.
5. Drop it on the chart or on a specific target that appears.

The available options depend on your chart type and the data you use.

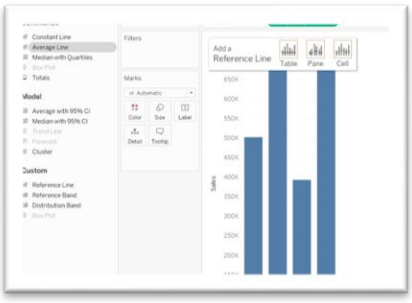
2. Important options in the Analytics pane



1. Constant Line

Draws a line at a fixed value that you choose.

Example: Add a line at ₹50,000 to see which products generate sales above or below your target.



2. Average Line

Shows the average value across your chart or a selected section.

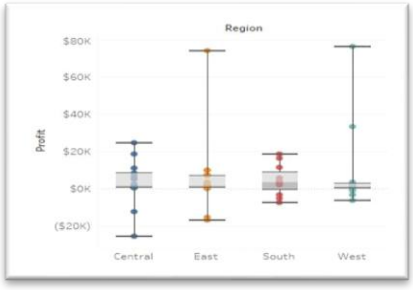
Example: If average monthly sales are ₹40,000, quickly identify months performing above or below average.



3. Median with Quartiles

Shows the median (middle value) and the quartiles that divide your data into four parts.

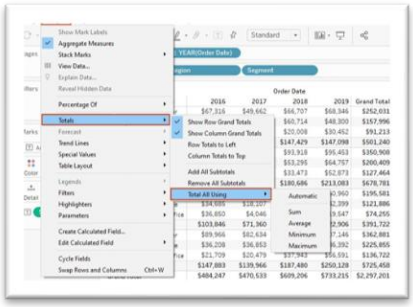
Example: Understand the typical student fee amount and how student fees are distributed.



4. Box Plot

Displays the median, quartiles, spread and potential outliers in your data.

Example: Compare the distribution of salaries across different departments.



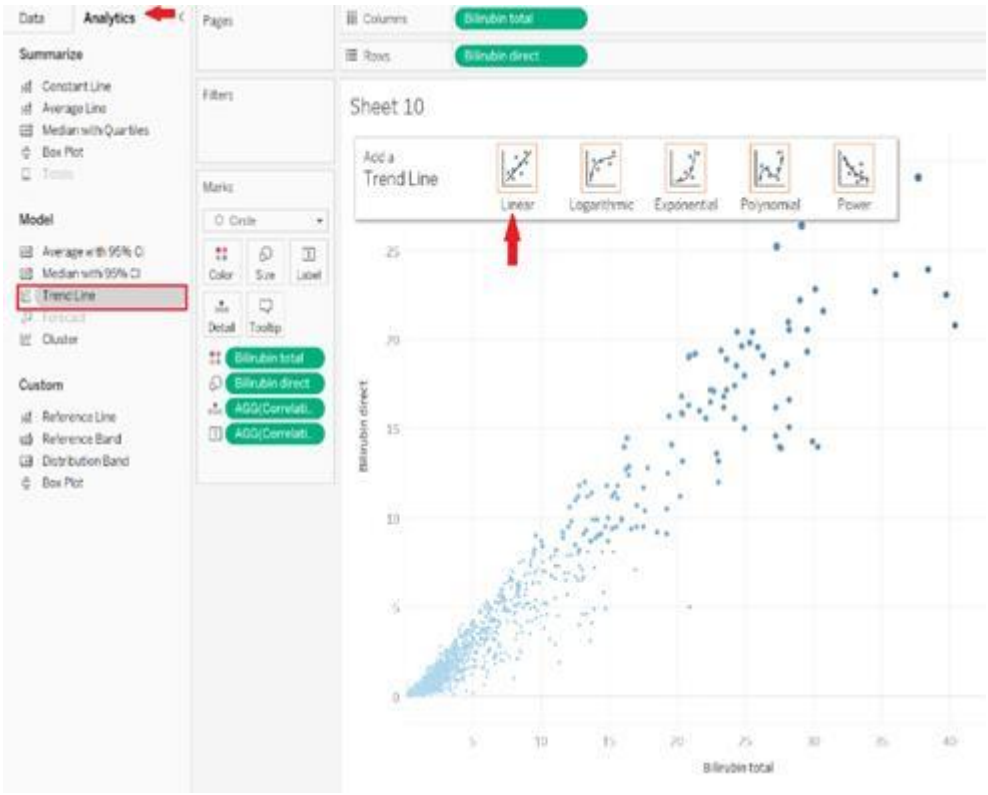
5. Totals

Adds subtotals or grand totals to a visualization.

Example: Show total sales for each region and the overall sales total.

These tools belong to the Summarize section.

3. Model section — understand patterns and make predictions



6. Trend Line

Identifies the general relationship between variables or how values change over time.

Example: Determine whether monthly sales are generally increasing or decreasing.

Common models: Linear, Logarithmic, Exponential and Polynomial.



7. Forecast

Estimates future values using historical data, particularly time-series data.

Example: Use previous months' sales to estimate sales for the next three months.

Forecasts are estimates, not guaranteed results.



8. Cluster

Groups similar data points based on selected variables.

Example: Group students according to attendance, course fees and performance to identify similar patterns.

This is useful for customer segmentation and exploratory analysis.

9. Average with 95% CI

Displays the average along with a 95% confidence interval, showing uncertainty around the estimated population average.

Example: Estimate the average examination score for a wider student population from a sample.

The Model section contains statistical and predictive tools. The exact options available depend on your visualization.

4. Custom section — create your own analytical reference points

10. Reference Line

Add a line based on a value or calculation you specify, such as a target, average or median.

11. Reference Band

Shades a region between two values.

Example: Highlight sales between ₹40,000 and ₹60,000 as your desired performance range.

12. Distribution Band

Adds a shaded area representing a statistical distribution or spread of values.

Example: See the range within which a large proportion of observations fall.

The Custom section lets you configure these elements rather than relying only on predefined analytical settings.

5. What do Table, Pane and Cell mean?

When you drag an item into your visualization, Tableau may show three drop targets.

Table

Applies the analysis across the entire table or visualization.

Pane

Applies the analysis separately to each section of the visualization.

Cell

Applies the analysis to an individual cell or mark, depending on the view.

For example, if you have sales by region and category:

- Table: One overall average sales line.
- Pane: A separate average for each region's pane.
- Cell: An average for each individual cell, where supported.

The exact result depends on how the view is arranged.

6. Practical exercise — try it in Tableau

Suppose you have this sample data:

Month	Sales (₹)
January	30,000
February	40,000
March	50,000
April	60,000
May	70,000

Experiment with Analytics

Select a tool to see what it helps you discover.

Average Line

Constant Line

Trend Line

Forecast

Cluster

Average sales: ₹50,000

January and February are below average, March equals the average, and April and May are above average.

To try this in Tableau, put Month on Columns, Sales on Rows, and choose a bar chart. Then open the Analytics pane and drag your selected tool into the view.

My recommendation: Learn Average Line, Reference Line, Trend Line, Forecast and Cluster first. These are especially useful when building business dashboards and performing data analysis.

For the official guide, see Tableau's Analytics Pane documentation.